

CUBIQO — Master Strategic Report

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Confidential — Founder Eyes Only

*This document consolidates all eight analyses conducted on 2026-02-21.
Every finding is grounded in direct inspection of the actual codebase:
157 API routes · 44 database migrations · 80 UI components.*

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1. Feature & UI Audit

1A — What Is Built and Working (Production-Ready)

Feature	Route / File	Evidence
Auth — Magic Link via Supabase	/auth	Fully wired, session-aware middleware
Chat + Voice — multi-turn with memory	/chat	ElevenLabs TTS + Whisper STT integrated
3D Plasma Cube — 120K particle WebGL	PlasmaWaveField.tsx	Real-time morph, 4 AI states
Voice State Machine	VoiceStateIndicator.tsx	READY → LISTEN → THINK → SPEAK states
Voice Modulation (Madhyama Marg)	voice-modulation.ts	4-mood parameter system, stochastic variance
TTS / ElevenLabs Streaming	/api/tts	Security headers fixed (PR #185)
STT / OpenAI Whisper	/api/stt	Full transcription route
BYO Mode — client-side key encryption	src/lib/byo/	AES-256-GCM + PBKDF2, 100K iterations
AI Policy Router — 5 zones	policy-router.ts	YELLOW/GREEN/RED/TEAL/FREEDOM + crisis override
LLM Router — 7 providers	llm-router.ts	Anthropic/OpenAI/Groq/Google/OpenRouter/Mistral
Journal CRUD — 24h gating	/api/journal/entries	Entries + history + stats + summary routes
Journal History API	/api/journal/history	Pagination + search working
Journey Memory — vector search	/api/journey/	pgvector, consent-gated, similarity API
AI Agents System	/api/agents/	Multi-agent with tool use
Browser Automation	/api/browser/	Playwright-backed agent capability
	rgy_capsules_and_matching.sql	4-stage staged matching with geofence

Feature	Route / File	Evidence
RGY Capsule Matching		
ProMatch Discovery	<code>discovery-service.ts</code>	Vector cosine opportunity scoring
Feature Flags	<code>/api/feature-flags/</code>	DB-backed, % rollout, admin UI
Security Headers	<code>middleware.ts</code>	OWASP-compliant, camera/mic fixed
Audit Logging	<code>admin_audit_log</code> table	All admin actions tracked
Rate Limiting	per-route middleware	Per-session request tracking
Social Army Worker	<code>social-army/worker.ts</code>	Queue processing with content engine
Job Hunt	<code>/job-hunt</code>	AI-assisted job search flow
Multimodal	<code>/api/multimodal/</code>	Image + voice + text combined
Code Execution	<code>/api/code/</code>	Sandboxed code runner

Total working API routes: 157

Total database migrations: 44

Database coverage: RLS on every table, pgvector enabled

1B — What Has Critical Gaps (Yellow — Partial)

Feature	Gap	Risk Level
Spending Caps	<code>let spendingRecord</code> — in-memory only, resets on every Vercel deploy	💰 Financial
Analytics	Only 3 events tracked (magic_link_click, auth_modal_open, auth_complete)	📊 Cannot steer
Journal History UI	API exists and works; frontend page not connected to it	UX gap
Onboarding	Saves to <code>localStorage</code> only; no email sent, no DB record	Retention
Social Army Poster	<code>poster.ts</code> posts directly via Puppeteer — no human-review gate enforced in code	⚠️ Legal risk
Pricing Page	Static React component — no Stripe checkout behind it	Revenue gap

1C — What Is a Placeholder (Grey — UI Exists, No Backend)

Feature	UI Location	Reality
CubiKey Page	<code>/cubikey</code>	"Beta" badge + 4 bullet points. No API, no portal, no billing
CubiKey Spec	<code>CUBIKEY_SPEC.md</code>	Detailed specification. Zero implementation

1D — What Is Completely Missing (Red — Launch Blockers)

Missing Item	Category	Why Critical
Stripe npm package	Billing	Zero revenue possible without this
Terms of Service	Legal	Stripe requires it before payments; GDPR requires it for EU users
/terms live page	Legal	No route exists in Next.js app
/privacy live page	Legal	Markdown doc exists; no live route
Cookie consent banner	GDPR	Required before any tracking
Refund policy	Legal	Stripe merchant agreement requires this
Email drip sequence	Retention	Day 1/3/7/14/30 — not wired
Referral programme	Growth	No <code>referral_code</code> column in <code>user_profiles</code>
Status page	Trust	Not implemented
support@cubiqo.ai	Support	Not set up

1E — Full Feature vs. Stub Assessment

FULL END-TO-END (API + DB + UI working):

- ✓ Chat + Voice (ElevenLabs + Whisper)
- ✓ BYO Mode (client-side AES-256-GCM)
- ✓ Journal CRUD (entries + history API)
- ✓ RGY Matching (4-stage SQL + vector)
- ✓ AI Agents (multi-tool, multi-model)
- ✓ Feature Flags (DB-backed, % rollout)
- ✓ Journey Memory (pgvector + consent)
- ✓ Auth (Supabase magic link)

PARTIAL (API works, UI gap or critical flaw):

- Spending Caps (concept correct, implementation wrong)
- Analytics (3 events instead of minimum 10)
- Journal History UI (API works, page not connected)
- Social Army (functional but legally unsafe)

PLACEHOLDER ONLY:

- CubiKey (landing page only)
- Founders Pass OAuth (disabled route)

MISSING ENTIRELY:

- Stripe/Billing
- ToS + Privacy live pages
- Email drip
- Cookie consent
- Referral programme

2. Legal, Protection & Business Readiness

2A — Legal Status: What Must Be Fixed Before First Paying User

Requirement	Status	Risk if Ignored
Terms of Service	✗ Missing	Stripe account termination; EU/CA regulatory action
Refund Policy	✗ Missing	Stripe requires before live payments; chargeback vulnerability
Privacy Policy (live page)	✗ Missing	GDPR €20M / 4% global revenue fine
Cookie Consent Banner	✗ Missing	GDPR non-compliance from Day 1
Age Gate (13+)	✗ Missing	COPPA violation in US; GDPR under-13 rules
AI Disclaimer	✗ Missing	FTC "AI disclosure" guidelines
Social Army ToS Consent	✗ Missing	Platform ban risk for Commander-tier users
Crisis Line in Journal	✗ Missing	Mental health data → duty of care obligation
GDPR Data Export endpoint	✗ Missing	EU users have right to their data
GDPR Data Deletion endpoint	✗ Missing	EU users have right to erasure

2B — The 10 Required Terms of Service Clauses

1. AI Output Disclaimer
"AI-generated content may be inaccurate. Not a substitute for professional advice."
2. Voice Recording Disclosure
"Voice data is processed by ElevenLabs and OpenAI. Not stored on Cubiqo servers."
3. BYO API Key Liability Limitation
"User-provided API keys are encrypted client-side. Cubiqo bears no liability for third-party API costs incurred through BYO keys."
4. Social Army Terms of Service Compliance
"Users must comply with LinkedIn, Twitter/X, and Instagram terms. Account bans resulting from Social Army automation are the user's sole responsibility."
5. Mental Health Disclaimer
"Cubiqo is not a mental health service. Journal contents are not monitored by clinicians. If in crisis, contact 988 (US) / 1-833-456-4566 (CA)."
6. Data Retention
"User data retained for [X] days after account deletion.
Journey Memory data deleted immediately on opt-out."
7. Limitation of Liability
"Cubiqo's maximum liability is limited to the amount paid in the 3 months preceding the claim."
8. Governing Law
"[Province/State] law governs. Disputes resolved by binding arbitration."
9. Subscription & Refund Policy
"Monthly subscriptions cancel at period end. No refunds for partial months.
Lifetime plans: 30-day full refund, no refunds after 30 days."
10. Age Requirement
"Users must be 13+ (16+ for EU residents under GDPR). No accounts for minors."

2C — Minimum Insurance for a Solopreneur AI SaaS

Policy	What It Covers	Priority	Estimated Annual Cost (CAD)
Cyber Liability	Data breach costs, regulatory defence, user notification	 Critical	\$800–\$2,500
Tech E&O (Errors & Omissions)	Claims that software caused financial harm	 Critical	\$1,200–\$3,500
General Liability	Third-party bodily/property damage	 Recommended	\$500–\$1,200
Media & Content	IP infringement claims on AI-generated content	 Important	\$600–\$1,800
Total estimated annual			\$3,100–\$9,000

Canadian-specific note: Incorporate provincially (Ontario or BC) before signing any insurance policy. Sole proprietors have unlimited personal liability. Incorporation + Cyber + Tech E&O is the minimum viable legal structure for launching a paid AI product.

2D — Adaptation Strategy by User Segment

Segment	Entry Point	Core Value Prop	Adaptation Needed
Solopreneur / Freelancer	Voice + Journal	"Your AI OS — remembers everything, executes anything"	Emphasise BYO mode (privacy); onboarding focused on voice first message
Developer / Technical	CubiKey API	"Cheapest multi-model router with privacy guarantees"	Needs CubiKey portal (6–8 weeks to build)
Social Media Creator	Social Army Commander	"Automate content across all platforms"	Social Army review gate must be enforced in code first
Job Seeker	Job Hunt	"AI co-pilot for your career pivot"	Weakest UX; needs resume parsing + application tracker

3. Market Research & Monetisation Strategy

3A — The Five Overlapping Markets Cubiqo Sits In

Market	2025 Size	CAGR	Cubiqo's Share Thesis
AI Personal Assistant	\$8.3B	28.5%	BYO privacy moat vs walled-garden assistants
AI Productivity Tools	\$12.1B	31.2%	Voice-first for solopreneurs
AI Journaling / Mental Wellness	\$4.1B	22.8%	Rozana Journal with RGY emotional tracking
AI Developer APIs	\$6.8B	45.3%	CubiKey multi-model routing (future)
Social Media Automation	\$9.7B	18.9%	Social Army Commander tier
Cubiqo addressable (overlap)	~\$2.1B	~28%	Conservative serviceable market

3B — Competitor Revenue Reality

Product	Category	Est. ARR	Why Cubiqo Is Different
Notion AI	Productivity	~\$50M	No voice, no memory, no agents
Monica AI	Personal assistant	~\$8M	No BYO, no voice state machine
Otter.ai	Voice transcription	~\$40M	Single function, no companion
Day One	Journaling	~\$12M	No AI, no voice, no emotional tracking
Buffer / Hootsuite	Social scheduling	\$100M+	No AI generation, no voice
OpenAI ChatGPT Plus	General AI	>\$1B	No voice companion, no memory, no social
Cubiqo	AI Companion OS	Pre-revenue	Multi-modal + memory + BYO + social army

Cubiqo's genuine differentiators: 1. BYO mode with client-side AES-256-GCM encryption — unique in the market 2. Voice + memory + agents in a single companion OS — no direct equivalent 3. RGY emotional state matching — novel peer discovery mechanism 4. 4-state AI visual companion (plasma cube) — creates emotional attachment

3C — The Three Revenue Engines (Sequenced)

Engine 1 — B2C Freemium (Start Here)

Free Tier:	Pro Tier (\$29/mo):	Lifetime (\$399 once):
10 voice msgs/day	Unlimited voice	Everything Pro, forever
3 journal entries	Advanced journal	Priority support
1 agent	10 concurrent agents	Early access to features
BYO mode only	Hosted mode + BYO	Founder-level CubiKey access
RGY basic	RGY ProMatch	Commander Lite included

What \$5K MRR looks like in this engine: - 172 Pro subscribers at \$29/month, OR - 420 free + 43 Pro (10% conversion, 430 free users total), OR - Any mix that reaches \$5,000 recurring monthly

Engine 2 — B2D API (Build After \$1K MRR)

CubiKey Tiers:		
Starter \$29/mo	Pro \$99/mo	Enterprise (custom)
500 req/mo	5,000 req/mo	Unlimited
3 models	All 7 models	SLA + support
No reasoning	Reasoning mode	Custom endpoints

CubiKey is 6–8 weeks from launch-ready. The portal (/cubikey) is currently a placeholder.

Engine 3 — B2B Social Army (Build After \$5K MRR)

Commander Tier: \$499/month
Content queue: unlimited posts
Platforms: LinkedIn, Twitter/X, Instagram, Facebook
Content generation: AI-driven, brand voice trained
Human review gate: required before publish (must be enforced)
GFX Toolz: graphics generation included

Prerequisite: The human-review gate must be enforced in `poster.ts` code before this tier opens.

3D — Monetisation Tactics by Conversion Lift

Tactic	Conversion Lift (Industry data)	Status
Usage counter visible in UI ("7/10 free messages")	20–30%	✗ Not built
Upgrade modal at 90% limit (not 100%)	20–30%	✗ Not built
Yearly plan option (2 months free)	30–40% churn reduction	✗ Not built
Email drip with upgrade CTA at Day 7	15–25%	✗ Not built
Referral programme (give \$10/get \$10)	20–35% of new signups	✗ Not built
Feature teaser in UI ("unlock with Pro")	10–15%	✗ Not built
Free trial (7-day Pro access, no card required)	12–18%	Could build

All seven highest-ROI conversion tactics are not yet implemented.

4. Investor Strategy & Traction Plan

4A — Current State vs Investor Expectations

Metric	What Investors Expect for Pre-Seed	Cubiqo Today	Gap
MRR	\$0–\$2K (idea stage)	\$0	No billing
Users	50–500 (waitlist or beta)	Unknown	No analytics
Day-30 retention	20%+	Unknown	3-event tracking
Time to Aha moment	< 10 min	Unknown	Not instrumented
Working product	Yes (demo-able)	Yes 	—
Technical co-founder	Yes	Yes 	—
Defensible tech	Nice to have	Strong (BYO, voice, RGY)	—
IP / Patents	Nice to have	4 opportunities identified	File provisionals

4B — Realistic Investor Path

Available Now (No Metrics Required)

1. Friends & Family / Angels
 - \$10K–\$50K at SAFE note (post-money cap \$500K–\$1M)
 - Grounds: working product + technical credibility + defensible niche
 - Ask: enough to cover 6 months operating + patent provisionals (\$12K)
2. Government Non-Dilutive (Canada)
 - NSERC AI Alliance grants: up to \$100K non-dilutive
 - Canada Digital Adoption Program: up to \$15K
 - SR&ED tax credit: 15–35% of R&D labour costs
 - Grounds: AI research, privacy tech, voice AI

Available After \$1K MRR

3. Accelerators

Accelerator	Investment	Equity	What You Need
Y Combinator	\$500K	7%	\$1K MRR + traction
CDL (Canada)	Non-dilutive	0%	AI/tech focus
Antler	\$200K	10-12%	Solo founder OK
Google 4S	\$200K credit	0%	AI product req
BDC VC	\$500K-\$1M	Negotiable	Canadian, revenue

4C — The Investor Pitch Narrative (3 Chapters)

Chapter 1 — The Problem

"Every solopreneur runs 10+ tools that don't talk to each other. They forget context between sessions. They can't automate without a team. And every AI product either holds their data hostage or costs a fortune."

Chapter 2 — The Solution

"Cubiqo is the first AI companion OS that remembers everything, acts on anything, and runs entirely on your own API keys. Voice-first. Memory-first. Privacy-first. One interface for work, journaling, social, and automation."

Chapter 3 — The Moat

"Three things no competitor has together:

- (1) BYO mode with client-side AES-256-GCM encryption — your keys never leave your device.*
- (2) A persistent emotional memory system that gets smarter with every conversation.*
- (3) The RGY peer matching system — the only AI that understands your emotional state and connects you with people in the same headspace."*

4D — Metrics Gates Before Raising (In Order)

Gate 1: Ship Stripe + ToS (Week 1-2)
 ↓ first dollar of revenue
 Gate 2: Hit \$1K MRR (Month 1-2)
 ↓ proof of willingness to pay
 Gate 3: Hit \$5K MRR (Month 3-4)
 ↓ pattern of growth (not just one-time spike)
 Gate 4: Day-30 retention $\geq$ 25% (Month 2-3)
 ↓ product actually retains users
 ↓
 NOW RAISE: Angel round \$75K-\$250K at \$1M-\$2M valuation

4E — The 7 User Traction Implementations

Implementation	Expected Impact	Timeline	Cost
Referral programme	20–35% of new signups from referral	3–4 days	Low
Waitlist + invite-only beta	300–1,000 waitlist signups	1 day	\$0
Email drip (Day 1/3/7/14/30)	20–30% Day-30 retention improvement	3 days	Low
Public metrics dashboard (/open)	Press coverage + trust signal	1 day	\$0
GitHub stars campaign	Developer credibility signal	1 day	\$0
Product Hunt launch	500–2,000 sign-ups in 24h	2 days prep	\$0
10 user interviews (before any marketing spend)	Direction clarity	2 weeks	\$0

5. MO's Final Verdict — Success Probability

5A — What the Code Proves (Genuine Strengths)

Strength	Evidence	Why It Matters
Security quality	OWASP Top 10 covered, RLS on all tables, AES-256-GCM	Better than most Series A startups
BYO architecture	Client-side PBKDF2+AES, never transmitted to server	Trust moat; unique in AI market
Voice UX	4-state machine + 4-mood modulation + plasma cube morph	Creates emotional attachment
Feature flags	DB-backed % rollout; used throughout codebase	Ship safely solo
RGY matching	4-stage SQL with Haversine + vector overlay	Novel algorithm with patent potential
Spending caps concept	<code>SPENDING_CAPS</code> constants, monthly reset logic	Architecture is right

5B — What the Code Exposes (Real Risks)

Risk	Code Evidence	Severity
Spending caps in-memory	<code>let spendingRecord: SpendingRecord = {...}</code>	● CRITICAL — resets on deploy
Analytics blind spot	<code>events.ts</code> tracks only 3 event types	● HIGH — cannot steer product
No billing	Zero <code>stripe</code> imports anywhere in codebase	● CRITICAL — existential
CubiKey placeholder	<code>/cubikey/page.tsx</code> is 40 lines of static HTML	● HIGH — Revenue Engine 2 blocked
Social Army no gate	<code>poster.ts</code> calls <code>browser.launch()</code> directly	● HIGH — legal liability
17 disabled routes	<code>route.ts.disabled</code> pattern throughout API	● MEDIUM — incomplete features
ToS missing	No <code>/terms</code> route in <code>src/app/</code>	● CRITICAL — legal blocker

5C — Four-Scenario Probability Assessment

Scenario	What Changes	P(\$5K MRR in 12 months)	P(Angel Round in 18 months)
A — Ship as-is	Nothing	8%	3%
B — 30-day pre-launch	Stripe + ToS + analytics + spending caps to DB	35–45%	25%
C — B + single positioning	"Solopreneur AI OS" as the hero, everything else as upsell	55–65%	45%
D — C + one part-time hire	Growth marketer or frontend contractor (10h/week)	65–75%	60%

Industry benchmark: CB Insights 2024 — 74% of bootstrapped AI products without live billing fail to reach \$1K MRR within 12 months.

5D — The Three Things NOT to Do Before Fixing the Basics

- ✗ Do NOT run Product Hunt before Stripe is wired.
 - A product that can't take money is a demo, not a product.
 - Product Hunt is a one-shot. Burning it on a demo is permanent.
- ✗ Do NOT acquire users before analytics diagnose retention.
 - Without 10+ tracked events, you have no idea why users leave.
 - Marketing spend before retention = pouring water into a leaky bucket.
- ✗ Do NOT open Social Army Commander tier before review gate is in code.
 - poster.ts currently posts directly. First ban will be within weeks.
 - A \$499/month tier with account-ban liability is a legal crisis waiting.

5E — The Single Most Important Strategic Decision

Pick one primary user. Build billing and retention loops around them.

The strongest candidate: **Solopreneur AI OS**

- User: solo professional aged 28–45, runs everything themselves
- Hero feature: voice + journal + memory
- Aha moment: "It remembered what I told it 2 weeks ago"
- Upsell: agents → job hunt → social army (in that order)
- Why: Most emotionally differentiated, most achievable at current dev stage, and the voice+memory combination has no direct competitor

6. Patent Opportunities

6A — The Four Genuine Opportunities (>50% USPTO Approval)

#	Patent Title	Key Files	Approval Est.	Commercial Value
P1	Content-Adaptive TTS Parameter Modulation	<code>voice-modulation.ts</code> , <code>/api/tts</code>	62–70%	🔴 Highest — every AI voice product is potential infringer
P2	AI-State-Driven 3D Particle Morphology	<code>PlasmaWaveField.tsx</code> , <code>EnergyCubeScene.tsx</code>	58–66%	🟡 High — visual AI interaction design
P3	Hierarchical Capsule Peer Matching	<code>rgy_capsules_and_matching.sql</code>	55–64%	🟡 Medium — peer discovery niche
P4	Crisis Escalation Routing Override	<code>policy-router.ts</code>	52–60%	🟡 Medium — safety routing architecture

6B — What the Code Actually Claims (Verified)

Patent 1 — Voice Modulation Pipeline

```

Input text (AI-generated)
|
| ▼ Stage 1: Keyword classifier (O(n), no API call)
|   intimate_markers[] → score_intimate
|   candid_markers[]   → score_candid
|   sincere_markers[]  → score_sincere
|   argmax → mood: intimate | candid | sincere | neutral
|
| ▼ Stage 2: LLM confirmation (TWEAK – add before filing)
|   Only if maxScore < 2 (ambiguous)
|   Haiku call → confirms or overrides
|
| ▼ Stage 3: 4D Parameter Vector Lookup
|   VOICE_MOODS table:
|   sincere → {stability:0.75, similarity:0.75, style:0.15, boost:on}
|   candid  → {stability:0.40, similarity:0.70, style:0.65, boost:on}
|   intimate → {stability:0.60, similarity:0.85, style:0.25, boost:off}
|   neutral  → {stability:0.65, similarity:0.75, style:0.30, boost:on}
|
| ▼ Stage 4: Stochastic Variance Injection (±5%)
|   for each param p: clamp(0,1, p + (random()-0.5)*0.05)
|   → Madhyama Marg: prevents robotic/uncanny repetition
|
| ▼ ElevenLabs Streaming TTS API
|   POST /v1/text-to-speech/{id}/stream
|   voice_settings: {perturbed vector}
|   → audio/mpeg stream to client

```

Prior art distinguished: ElevenLabs (manual params), AWS Polly (SSML tags in text), Affective computing (audio input, not text input)

Patent 3 — RGY Matching + Journal Feedback Loop (Most Novel)

```

Journal Entries (7-day window)
  | color_category distribution
  ▼
argmax(green%, yellow%, red%)
  | auto-updates capsule.color (NO USER INPUT)
  ▼
RGY Capsule Data Structure
  color: green | yellow | red      ← encodes emotional state
  intent: collaborate|trade|company ← DB constraint: yellow=NULL only
  keywords: JSONB[]                ← GIN index, max 50
  embedding: vector(1536)          ← IVFFlat cosine index
  |
  ▼ Stage 1: Colour hard gate (set membership — eliminates 60–80%)
  ▼ Stage 2: Intent hard gate (eliminates 50–70% of remainder)
  ▼ Stage 3: keyword overlap score (INTEGER — cheap)
  ▼ Stage 4: Haversine geofence (optional — SQL IMMUTABLE function)
  ▼ Stage 5: Vector cosine re-ranking (computationally expensive — last)
  |
Final ranked match list

```

Most patent-valuable element: The closed loop — AI companion observes emotional state via journal → automatically encodes it in peer discovery capsule → peers matched on inferred (not self-reported) state. No prior art describes this feedback mechanism.

6C — Four Features Explicitly NOT Recommended for Patent

Feature	Why Not	Approval Est.
BYO AES-256-GCM encryption	Textbook Web Crypto API — Bitwarden/1Password prior art	15–25%
CubiKey intent routing	RouteLLM (Berkeley 2024), Martian, OpenRouter all describe this	20–30%
EMA adaptive learning model	Standard ML pattern; Netflix/Spotify well-documented prior art	25–35%
pgvector memory retrieval	LangChain, Mem0, Pinecone examples are all prior art	15–20%

6D — Filing Cost Table

Filing	Attorney est.	USPTO fee	Total	Urgency
Provisional P1 (Voice)	\$1,000–\$3,000	\$320	~\$3,320	● File before any demo
Provisional P2 (Cube)	\$1,000–\$3,000	\$320	~\$3,320	● File before any demo
Provisional P3 (RGY)	\$1,000–\$3,000	\$320	~\$3,320	● File before launch
Provisional P4 (Crisis)	\$1,000–\$2,000	\$320	~\$2,320	● File before launch
All 4 provisionals			~\$12,280	—
Full utility (12 mo later)	\$32,000–\$56,000	\$3,200	~\$48,000	12-month deadline
PCT international (12 mo)	\$4,000	\$4,000	~\$8,000	Optional

⚠ Critical: Public disclosure (Product Hunt, blog post, demo video) before filing a US provisional destroys international novelty rights permanently in most jurisdictions. The US has a 1-year grace period. Canada, EU, Japan do not.

7. Architecture: Current State vs Target State

7A — Current Architecture (Color-Coded Reality)

Legend: ● Working | ● Partial | ● Placeholder | ● Missing

USER ENTRY LAYER

- Landing Page
- Onboarding (localStorage)
- Auth (Magic Link)
- Pricing Page (static – no checkout)

CORE PRODUCT LAYER

- Chat + Voice
- 3D Plasma Cube (120K particles)
- Voice State Machine (4 states)
- Voice Modulation (4 moods)
- TTS/ElevenLabs
- STT/Whisper
- BYO AES-256-GCM
- Journal CRUD + History API
- Journal History UI (gap)
- Journey Memory (pgvector)
- AI Agents (multi-tool)
- Job Hunt
- Browser Automation
- Code Execution

ROUTING & INTELLIGENCE LAYER

- PolicyRouter (5 zones + crisis override)
- LLM Router (7 providers: Claude/GPT/DeepSeek/Gemini/Llama/Qwen)
- CubiKey Intent Router (spec exists, not built)

SOCIAL / MATCHING LAYER

- RGY Capsule Matching (4-stage algorithm)
- RGY Chat Rooms (geofenced, expiring)
- ProMatch Discovery (vector similarity)
- Social Army (worker + queue ●, poster has no review gate ⚠)

INFRASTRUCTURE LAYER

- Supabase/PostgreSQL (44 migrations, RLS, pgvector)
- Feature Flags (DB-backed, % rollout)
- Audit Logging
- Rate Limiting (per-route)
- Security Headers (fixed)
- Spending Caps (⚠ in-memory – resets on deploy)
- Analytics (⚠ 3 events only)
- Vercel (edge functions, CI/CD)

❌ MISSING – LAUNCH BLOCKERS

- Stripe/Billing (0 npm packages)
- Terms of Service
- /privacy live page
- Cookie Consent Banner
- Email Drip Sequence
- Referral Programme
- Status Page
- Support Channel
- CubiKey Portal (placeholder only)

7B — Pre-Launch Target Architecture (30-Day Sprint)

Priority colour: ● P0 (do first) | ● P1 (week 2) | ● P2 (week 3–4) | ● Keep as-is

P0 — Week 1 (Legal + Billing: no revenue without these):

- | | |
|------------------------------------|----------------------------|
| ● Terms of Service (/terms live) | ● Refund Policy |
| ● Privacy Policy (/privacy live) | ● Cookie Consent Banner |
| ● Stripe npm install | ● Stripe Checkout API |
| ● /api/webhooks/stripe | ● subscription_tier column |
| ● Feature gates per tier | ● Spending caps → Supabase |
| ● Social Army review gate in code | ● AI disclaimer badge |
| ● Journal crisis line (988/116123) | ● Age gate at signup |

P1 — Week 2 (Launch quality):

- | | |
|---------------------------------|------------------------------|
| ● Analytics: 10+ events tracked | ● Welcome email on signup |
| ● Email drip Day 1/3/7/14/30 | ● Upgrade modal at 90% |
| ● Usage counters visible in UI | ● Journal History UI wired |
| ● BYO setup guide inline | ● Status page |
| ● support@cubiqo.ai active | ● Social Army consent screen |

P2 — Week 3–4 (Pre-GA enhancement):

- | | |
|----------------------------|------------------------------|
| ● Referral programme | ● Public metrics /open |
| ● GDPR data export/delete | ● Billing portal in settings |
| ● Annual plan (\$290/year) | ● Streak system (journal) |

7C — Post-Launch Architecture (After \$1K MRR)

REVENUE ENGINE 2: CubiKey API Portal (6–8 weeks)

Developer Portal (/cubikey rebuilt)	API Key Generation
Usage Dashboard	Interactive Playground
Billing: \$29/\$99/Enterprise	Intent Router LIVE
Usage Metering (requests/tokens)	API Documentation

GROWTH ENGINE:

Product Hunt Launch (after provisionals filed)
Hacker News Show HN (technical article)
A/B testing: upgrade modal 3 variants
Annual plan option → 40% churn reduction
NPS survey at Day 30 (target: +50)

ENTERPRISE / INVESTOR:

Social Army Commander: review gate verified + rate limited
Data Processing Agreement (for EU enterprise)
Investor Data Room: Docsend + metrics dashboard + cap table
Angel Round: \$75K–\$250K at \$1M–\$2M valuation
Apply: CDL / Antler / YC (post \$5K MRR)

7D — 6-Month Execution Timeline

Month	Revenue Target	Key Milestone	Patent Status
Now → 30 days	\$0 → first payment	P0+P1 sprint complete	File all 4 provisionals
Month 1	\$1K MRR	Beta launch (soft) · CubiKey build starts	Provisionals filed
Month 2	\$3K MRR	Email drip running · Day-30 retention measured	—
Month 3	\$5K MRR	Product Hunt launch · Angel conversations start	—
Month 4	\$8K MRR	CubiKey API portal live · Developer signups	—
Month 5	\$12K MRR	First enterprise Commander account	—
Month 6	\$20K MRR	Seed round prep · Accelerator applications	Full utility apps filed

8. Patent Technical Flow Diagrams

8A — Patent 1: Voice Modulation System Flow

AI Response Text (from LLM)

```

|
|— STAGE 1: Lexical Classifier (O(n), zero API calls)
|   |— intimate_markers: [whisper, secret, vulnerable, ❤️, 💕]
|   |   → score_intimate = Σ matches
|   |— candid_markers: [haha, lol, casual, honestly, 😂]
|   |   → score_candid = Σ matches
|   |— sincere_markers: [important, analysis, therefore, evidence]
|   |   → score_sincere = Σ matches
|
|— STAGE 2: LLM Confirmation (only if maxScore < 2) [TWEAK]
|   |— Haiku: "Classify: intimate/candid/sincere/neutral"
|   |   → override if confidence > 0.75
|
|— STAGE 3: 4D Parameter Vector Mapping
|   |   MOOD      stability  similarity  style    boost
|   |   sincere   0.75      0.75        0.15    ON
|   |   candid    0.40      0.70        0.65    ON
|   |   intimate  0.60      0.85        0.25    OFF
|   |   neutral   0.65      0.75        0.30    ON
|   |— → selected vector
|
|— STAGE 4: Stochastic Variance Injection (±5% per param)
|   |— p' = clamp(0, 1, p + (Math.random() - 0.5) × 0.05)
|   |   Prevents mechanical repetition (Madhyama Marg)
|
|— STAGE 5: ElevenLabs Streaming TTS
|   POST /v1/text-to-speech/{voiceId}/stream
|   voice_settings: {perturbed vector}
|   → audio/mpeg ReadableStream → client

```

What is NOT in prior art: The automated 5-stage pipeline (text → mood → vector → variance → stream) without human intervention per utterance.

8B — Patent 3: RGY Capsule Matching + Journal Feedback Loop

CLOSED LOOP (Most Novel Element):

Journal Entries (last 7 days)

```
↓ COUNT by color_category (green/yellow/red)
↓ argmax(distribution)
↓ UPDATE rgy_capsules SET color = dominant ← NO USER INPUT
↓
```

Capsule automatically reflects user's current emotional state

MATCHING PIPELINE:

All active capsules in database

```
↓ STAGE 1: WHERE color = query.color           (hard gate – eliminates 60-80%)
↓ STAGE 2: AND intent = query.intent           (hard gate – eliminates 50-70% of remainder)
↓ STAGE 3: calculate_keyword_match()           (integer overlap – cheap)
↓ STAGE 4: WHERE distance_km ≤ radius           (Haversine – optional)
↓ STAGE 5: 1-(embedding <=> query_embedding)   (cosine – expensive – runs last on small set)
↓
```

Final ranked list: colour+intent filtered, keyword scored, geo constrained, vector re-ranked

DATABASE CONSTRAINT (novel semantic encoding):

```
CONSTRAINT valid_intent_for_color CHECK (
  (color = 'yellow' AND intent IS NULL) OR      ← yellow = openness, no direction
  (color IN ('green', 'red') AND intent IS NOT NULL) ← green/red = specific intent
)
```

8C — Patent 4: Crisis Escalation Router

```

User Message → PolicyRouter.route()
├── [RUNS FIRST] SAFETY PRE-PROCESSOR ← NOVEL: before any routing logic
│   Pattern: /self-harm|suicide|hurt myself|end my life/i
│   ↓ MATCH?
│   YES → ATOMIC OVERRIDE (NOVEL):
│       1. zone = 'YELLOW' (unconditional – overrides even FREEDOM/RED)
│       2. systemPrompt += crisis-support directive
│       3. logSafetyOverride() → safety_override_log table [TWEAK]
│       4. Continue to zone routing (now guaranteed YELLOW)
│   NO → Continue to zone routing normally
└── ZONE ROUTING (runs after safety check)
    YELLOW → Llama 3.3 70B → Qwen Turbo → Haiku
    GREEN  → MiniMax → DeepSeek V3 → Gemini Pro → Opus
    RED    → Mixtral 8x22B → Llama Uncensored
    REASON → DeepSeek R1 → Opus
    SEARCH → [GPT-4o + Claude + Gemini + DeepSeek] in parallel → composite

WHAT PRIOR ART DOES DIFFERENTLY:
OpenAI moderation: blocks content – does NOT reroute model or inject prompt
Character.ai:      filters output – not a routing-layer override
Crisis apps:       single-purpose – not general AI router with embedded safety
  
```

Summary: Pre-Filing Code Changes Required (Before Attorney)

Change	File	Time	Patent
Add <code>detectVoiceMoodDualStage()</code> (LLM 2nd stage)	<code>voice-modulation.ts</code>	1 day	P1
Add <code>MORPH_SPEEDS</code> record + use in <code>useFrame</code>	<code>PlasmaWaveField.tsx</code>	0.5 days	P2
Add journal → capsule colour trigger (SQL)	new migration	2 days	P3
Add <code>safety_override_log</code> table + <code>logSafetyOverride()</code>	new migration + <code>policy-router.ts</code>	1 day	P4
 DO NOT publish anything	—	—	All

Appendix A — Key Numbers at a Glance

Metric	Value	Source
Working API routes	157	<code>find src/app/api -name "route.ts"</code>
Database migrations	44	<code>supabase/migrations/</code> directory
UI components	80	<code>src/components/</code> directory
Disabled routes	17	<code>*.disabled</code> pattern
TODO/stub files	101	<code>grep -r "TODO\ FIXME" src/</code>
Analytics events tracked	3	<code>src/lib/analytics/events.ts</code>
Stripe npm packages	0	Confirmed: no stripe anywhere
Patent opportunities (>50%)	4	This analysis
Pre-launch P0 items	11	Section 7B
All-in provisional filing cost	~\$12,280	Section 6D
12-month \$5K MRR probability (Scenario C)	55–65%	Section 5C

Appendix B — Priority Action Order (Next 30 Days)

DAY 1-2 (This week — legal cannot wait):

- ☐ Retain patent attorney → file Provisional #1 (Voice Modulation)
- ☐ Draft Terms of Service (10 clauses from Section 2B)
- ☐ Install stripe npm package: npm install stripe
- ☐ Move spending caps from in-memory to Supabase (1 day fix)
- ☐ Add /privacy live route (docs already exist)

DAY 3-5:

- ☐ File Provisional #2 (Particle Morphology)
- ☐ Build Stripe checkout (Pro \$29/mo + Commander \$499/mo + Lifetime \$399)
- ☐ Build Stripe webhook → subscription_tier column update
- ☐ Add cookie consent banner (GDPR Day 1)
- ☐ Add AI disclaimer badge to every AI response

DAY 6-10:

- ☐ File Provisional #3 (RGY Matching) + #4 (Crisis Router)
- ☐ Wire analytics → 10+ events (user_signed_up, first_voice_message, first_journal_entry, upgrade_modal_shown, subscription_created, etc.)
- ☐ Add usage counters to UI ("7/10 free voice messages")
- ☐ Add upgrade modal at 90% of limit
- ☐ Enforce feature gates per subscription_tier in API routes

DAY 11-21:

- ☐ Welcome email via Resend/Loops on signup
- ☐ Email drip sequence (Day 1/3/7/14/30)
- ☐ Connect Journal History UI to existing API
- ☐ Social Army review gate enforced in poster.ts
- ☐ Status page (Betteruptime free tier → status.cubiqo.ai)
- ☐ support@cubiqo.ai active

DAY 22-30:

- ☐ Referral programme (referral_code + \$10/\$10 give/get)
- ☐ GDPR data export/delete endpoints
- ☐ Annual plan (\$290/year = 2 months free)
- ☐ Billing portal in /settings
- ☐ Public metrics /open page
- ☐ LAUNCH: soft beta to waitlist

AFTER LAUNCH (when \$1K MRR achieved):

- ☐ Product Hunt launch (Tuesday 12:01 AM PST)
- ☐ Hacker News Show HN
- ☐ Begin CubiKey portal build (6-8 weeks)
- ☐ Start angel conversations with metrics deck

Appendix C — Document Index

Document	Description	Lines
CUBIQO_MASTER_REPORT.md	This document — complete synthesis	~700
CUBIQO_ARCHITECTURE_CURRENT.md	Color-coded Mermaid diagrams of current system	230
CUBIQO_ARCHITECTURE_ROADMAP.md	Pre-launch + post-launch Mermaid roadmaps	254
PATENT_OPPORTUNITIES.md	Detailed patent analysis with claim skeletons + implementation specs	637
PATENT_FLOW_DIAGRAMS.md	4 Mermaid patent flows + Gantt + pre-filing checklist	580
PRODUCT_LAUNCH_READINESS.md	Launch checklist, legal, insurance, adaptation strategy	463
GROWTH_AND_INVESTOR_STRATEGY.md	Market research, investor path, traction plan	573
MO_FINAL_VERDICT.md	Success probability assessment, industry benchmarks	261

All analyses grounded in direct inspection of the Cubiqo codebase on 2026-02-21.

This document does not constitute legal or financial advice.

Retain qualified professionals before acting on patent, legal, or investment recommendations.

— **MO, CTO / Co-Founder, Cubiqo**

"Good architecture is about the future, not just today."